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Task 1 → Vector Representation

Numeber of student :- 5000
number of subject :- 15
complete score matrix shape = (5000, 15)
one student's vector shape (15)

L1 Norm :- vector ના બધા elements ના absolute value's addition

$$L1 = \sum_{i=1}^{n} |xi|$$

L2 norm :- vector ના બધા elements ના square ના સરવાળા ની square root લેવા.

$$L2 = \sqrt{\sum_{i=1}^{n} x_i^2}$$

L1 norm total magnitude
L2 norm vector's Eulidean magnitude

vector =

V = {83, 49, 75, 59, 46, 63, 55, 72, 51, 84, 66, 38, 48, 32, 52}

L1 norm cal =
L1 = 83 + 49 + 75 + ... + 52 = 896

L2 norm

$$L2 = \sqrt{83^2 + 49^2 + \dots + 52^2}$$

$$= \sqrt{55183} = 234.91$$

Dot Product

Dot product એટલે બે vector ના corresponding element ને multiply કરી ને મળતા product ના સરવાળા.

$$A \cdot B = \sum_{i=1}^{n} A_i B_i$$

$$= 69,939$$

Angle between two vector

બે vector વચ્ચે ના Angle શોધવા

$$\cos\theta = \frac{A \cdot B}{\|A\| \|B\|}$$

$$\theta = \cos^{-1}\left(\frac{A \cdot B}{\|A\| \|B\|}\right)$$

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Student A DSAI 0001
A = [83, 49, 75, 59, 46, 63, 55, 79 ... 52]

Student B DSAI 0002
B = [91, 82, 82, 87, 80 .... 70, 84, 88]

$$A.B = \sum_{i=1}^{15} A_i B_i$$

A.B = 7553 + 4018 + 6150 .... 3360 + 4452 + 4576
= 69,939

L2 norm

$$\|A\| = \sqrt{\sum A_i^2}$$

$= 83^2 + 49^2 + 75^2 + 59^2 .... + 48^2 + 55^2 + 52^2$
= 55,184

$\|A\| \sqrt{55,184}$

= 234.91

$\sum B_i^2 = 91^2 + 82^2 + 82^2 + 87^2 .... + 70^2 + 84^2 + 88^2$
= 91,752
$= \sqrt{91,752}$

= 301.91

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Angle between vector calculate

$$\cos\theta = \frac{A \cdot B}{\|A\| \|B\|}$$

$$\cos\theta = \frac{69,939}{234.912 \times 302.906}$$

$$= 0.98299$$

inverse cosine Apply

$$\theta = \cos^{-1}(0.98299)$$
$$= 10.61^{\circ}$$

Task 4 cross Product And Perpendi

Py:- first cordinate
Sql:- second cordinate
Excel:- third Cordinate

$$A \times B = \begin{bmatrix} A_2B_3 - A_3B_2 \\ A_3B_1 - A_1B_3 \\ A_1B_2 - A_2B_1 \end{bmatrix}$$

First: $A_2B_3 - A_3B_2$

$= (49 \times 82) - (75 \times 82)$

$= 4018 - 6150$

$= -2132$

Second $A_3B_1 - A_1B_3$

$(75 \times 91) - (83 \times 82)$

$= 6825 - 6806$

$= 19$

Third $= A_1B_2 - A_2B_1$

$= (83 \times 82) - (49 \times 91)$

$= 6806 - 4459$

$= 2347.$

Final cross product

$A \times B = [-2132, 19, 2347]$

~~Per~~ Dot Product :-

$= (83 \times -2132) + (49 \times 19) + (75 \times 2347)$

$= -176956 + 931 + 176025$

$= 0$

verification with Student B

$B = (91, 82, 82)$

$B \cdot (A \times B)$

$= (91 \times -2132) + (82 \times 19) + (82 \times 2347)$

$= -194012 + 1558 + 19454$

$= 0$

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Vector Projection

A = 83, 49, 75
B = 91, 82, 82

A.B = 17721
$||B||^2$ = 21729

$$\text{Proj}_B A = \frac{A.B}{||B||^2} B$$

$$= \frac{17721}{21729} (91, 82, 82)$$

$$= 0.81554$$

X = 0.81554 × 91 = 74.22
Y = 0.81554 × 82 = 66.82
Z = 0.81554 × 82 = 66.82.

Matrix creation

A = 83, 49
B = 91, 82

matrix A:

$$A \begin{bmatrix} 83 & 49 \\ 91 & 82 \end{bmatrix}$$

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Matrix addition

$A + A = 2A$

$$A + A = \begin{bmatrix} 83 + 83 & 49 + 49 \\ 91 + 91 & 82 + 82 \end{bmatrix}$$

$$= \begin{bmatrix} 166 & 98 \\ 182 & 164 \end{bmatrix}$$

Matrix multiplication

First Row = $(83 \times 83) + (49 \times 91) = 11341$

First row second column = $(83 * 49) + (49 \times 82) = 8085$

second row, first column = $(91 \times 83) + (82 \times 91) = 15015$

second row, second column = $(91 \times 49) + (82 \times 82) = 11183$

Final result

$$A^2 = \begin{bmatrix} 11341 & 8085 \\ 15015 & 11183 \end{bmatrix}$$

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Matrix Transpose

original $A = \begin{bmatrix} 83 & 49 \\ 91 & 82 \end{bmatrix}$

Transpose $A^T \begin{bmatrix} 83 & 91 \\ 49 & 82 \end{bmatrix}$

Determinant

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$|A| = ad - bc$$
$$= (83 \times 82) - (49 \times 91)$$
$$= 6806 - 4459$$
$$= 2347$$

matrix Inverse

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A^{-1} = \frac{1}{2347} \begin{bmatrix} 82 & -49 \\ -91 & 83 \end{bmatrix}$$

$$= \begin{bmatrix} 82/2347 & -49/2347 \\ -91/2347 & 83/2347 \end{bmatrix}$$

$$= \begin{bmatrix} 0.03494 & -0.02088 \\ -0.03877 & 0.03536 \end{bmatrix}$$

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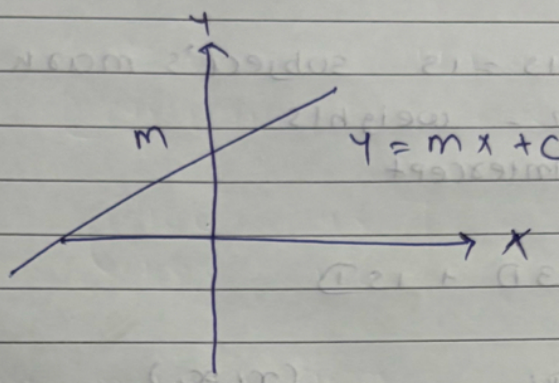
Task 1

Line - 2D Space



y = SQL marks
x = Python marks.

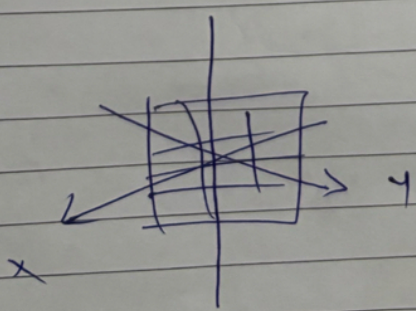
St 1 = 83, 49 St 2 = 91, 82	y = mx + c

Plane - 3D

x-axis = Python
y axis = SQL
z axis = Exce

St 1 = 83, 49, 75
St 2 = 91, 82, 82,

$ax + by + cz + d = 0$



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Hyper Plane

$w_1 x_1 + w_2 x_2 + \dots + w_{15} x_{15} + b = 0$

$x_1, x_2, \dots x_{15}$ = 15 subject's mark
$w_1, w_2, \dots w_2$ = weights
b = bias / intercept

2D → 3D → 15D

2D	PY, SQL	(x_1, x_2)
3D	PY, SQL, Excel	(x_1, x_2, x_3)
15D	15 subject	$(x_1, \dots x_{15})$

Task

$$cov(x, y) = \sum_{i=1}^{n} \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Python variance	323.65
SQL variance	332.02
covariance	232.98
covariance	227.52

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Task 9

Eigen value (λ)
Eigen vector (v)

	15 x 15
Covariance matrix	
Largest eigen value λ_1	3551.63
second eigen value λ_2	101.52
Third eigen value λ_3	101.17
Total variance	4869.09

$$\frac{3551.63}{4869.09} \times 100 = \boxed{72.94\%}$$